BTS-IV-04.18-0719 Reg. No. | | | | | | | | |

A

B.Tech. Degree IV Semester Examination April 2018

SE 15 - 1402 HEAT AND MASS TRANSFER OPERATIONS
(2015 Scheme)

Time: 3 Hours Maximum Marks: 60

PART A
(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) How does thermal conductivity for solids, liquids and gases depend upon temperature?

(b) How is heat transfer coefficient defined? What is the physical significance of 'film thicknesses? Explain Newton's law of cooling.

(c) How is absorptivity, reflectivity and transmissivity important in radiation?

(d) What is fouling factor? Discuss its importance in design of heat exchangers.

(e) For a solution, prove $D_{AB} = D_{BA}$.

(f) Obtain the expression for Ficks law and also mention the conditions under which it is valid.

(g) Obtain the expression for molar flux of liquids for steady state diffusion of A into non diffusing B.

(h) State Raoult's law and Henry's law and why they are significant.

(i) What is the physical significance of HTU and NTU?

(j) Draw and explain working of a bollmann extractor.

PART B

$(4 \times 10 = 40)$

II. (a) An annealing chamber has a composite wall made of a 17 cm thick firebrick layer (k = 1.1W/m°C) and a 13 cm thick ordinary brick layer (k = 0.70 W/m°C). The inside and outside surface temperatures of the walls are 400°C and 45°C respectively. Calculate the heat loss from 25 m^2 furnace wall. Also, determine the temperature between the ordinary brick and the fire brick layers. (5)

(b) A 1 kW electric room heater has a coil of Nichrome wire of diameter 0.574 mm and electrical resistance of 4.167 ohm/m. If the temperature of the room remains constant at 21°C and the average heat transfer coefficient at the wire surface is 100 W/m^2°C, calculate the time required for the heating coil after it is switched on to reach 63% of its steady state temperature rise. Assume that the wire offers negligible heat transfer resistance. The density of the material of the wire is 8920 kg/m^3, and its specific heat is 384 J/kg °C. (5)

OR

III. (a) State and discuss: (6)

(i) Reynolds Analogy.
(ii) Prandtl analogy.
(iii) Colburn analogy.

(b) Discuss the concept of boiling in heat transfer. Explain the various regimes involved. (4)

IV. What are the various factors involved in the design of heat exchangers? Draw the schematic of a single pass and multi pass heat exchanger. (10)

OR

V. (a) What is meant by radiation heat transfer? Discuss the concept of black body. Explain the laws of radiation. (6)

(b) Discuss the different types of evaporators and the method of feeding multiple effect evaporators. (4)

VI. Discuss the different theories of mass transfer. (10)

OR

VII. Design a packed tower with respect to: (10)

(i) Individual mass transfer coefficient.
(ii) Overall mass transfer coefficient.
(iii) Height of transfer units.

VIII. Explain and derive the McCabe Thiele method for design of a distillation column. (10)

OR

IX. Discuss: (10)

(i) Simple Leaching.
(ii) Sprays dryers and its industrial application.

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